



Dr.WASIMAKRAM BINNAL

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Total Experience: 15 year (12+ years in Data Science, Analytics and Data Engineering)

SUMMARY

A Data Science/Analytics professional with Doctorate in Consumer Analytics having around 15yrs of experience in developing projects across Retail, Fintech and Healthcare services and Information Technology domains. Focussed on developing solutions which are actionable, robust, and scalable. Experience in hands on end-to-end deployment of AI, GenAI & ML models in Cloud architecture like GCP & AWS. Developing Cloud architecture from scratch to scale micro-services and Analytics reporting.

Experienced in leading projects which involves more than 260+ clients across countries. Expertise in developing client relations, developing POC's for new solutions, Integration with legacy systems, Forecasting models, Deep learning models. Managed data science teams, MDM (Data Engineering), Model Validation and Data Analytics. Filed **four US patents** in domains of data engineering, inventory management, clinical predictions, and analytical based machine learning models.

EXPERIENCES

Headlamp Health Bengaluru

Aug 05, 2024 to Present

Role: Founding AI Architect

Data Science

Developing Analytical and AI solutions for Mental Health and Pharma. The focus is to develop products from scratch to increase patient engagement.

- Somatic Score: Developed a Prediction ML model (Random Forest Multi-Class Classifier) by using Apple Health Data. The predictions are for next 3 weeks performance of individuals and the top 2 factors are recommended to offset the behaviour. This involves pulling data from AWS S3 bucket JSON data. This data is normalised, standardised and aggregated for a daily basis per patient. The scoring formula is based on personal and industry benchmarks.
- Pharma ML: Prediction of which of the Apple Health metrics with medication is key in influencing patient moods. This utilises an MANOVA model for identifying the most influential medications. This involves pulling API data from RXNORM and matching the medications to establish a Medication Longitudinal record.

HST Global Hy-Vee, Inc. Bengaluru (Retail)

Aug 14, 2023 to Aug 02, 2024

Role: Senior Principal Data Scientist

Data Science

Establishing data science model architectures and production design for retailing and services. Managing a team of total 13 data engineers/analysts (MDM, Data Science and Analytics) for developing unique customer demographics for Hy-Vee. All the models are hosted in Google Cloud with end to end automation.

- Revamped MDM- Match and merge using **GenAI** and resulting in savings of **\$1.2Million** every year. Ramped off from Informatica and built a solution hand-on with accuracy of 98% using GenAI and embedding.
- Ramped of from H2O Data science platform to build everything in-house using GCP Vertex-AI resulting in saving of **\$240k** every year.
- Developed architecture for Text based Solutions for product categories auto tagging. Utilised Glove architecture with CNN model having 100D to result in accuracy for level1 category-99.7% & level3 category-97.3%.

- Software Test Case Generator: GENAI based LLM solution using llama-3 (7B Q5 version), LangChain with RAG based word embedding model. We auto feed any pdf, .txt or html page as input to generate an Embedding for specific software product (daily) and then any prompts can be passed using VPN based web-page for UI.
- Developed Regression and Classification prediction model for churn customer (attrition), accurate sales based regression model which forecast demand of items and seasonal items sales. A regressive model also deployed for commissary items with automated seasonal demand scaling.
- Image processing: Using GenAI models to tag, describe, classify and match images to product categories/items using Mistral, Llama & Dolphin with RAG. Hosting within the cloud using Apache beam, Spark architecture.
- Attribution Model: Developed channel attribution modelling using Bayesian Network to attribute weightage for different marketing channels for specific product items and categories.

Oracle Cerner Healthcare Solutions India Pvt Ltd, Bangalore

Jan 06, 2020 to Aug 10 2023

Role: Senior Manager Data Science

Data Science

Responsible for leading a team of 8 Data-Analyst/Data-Scientists who work on diverse projects ranging from Image-processing, NLP, and ML models for **260+ clients**. I primarily establish **analytical system designs** with focus on utilising **ML/DL** for business solutions. This includes end to end development to deployment of solutions with strategy to utilise machine learning outcomes. We monitor all models and solutions of ML/analytics using dashboards. Hands on approach for new ML model solutions to establish a standard template for the team. All ML solutions have real –time accuracy of 95+ percent. The projects details are,

- Deep learning and GPT based models for (a) Emergency registration (b) Capacity management (c) resource utilisation prediction (d) Visitation prediction models for healthcare management (d) Developed GPT-3 based Master patient record matching solution
- NLP Based projects: (a) Developing an engine for Data Engineering to convert text to numeric (b) Predictive model for accuracy of conversion (c) Classification model for feature identification (d) Recommendation engine for Jira/Remedy tickets
- Clinical Usecases: (a) Developed predictive model for assessing risk of diabetes during pregnancy, (b) Predictive model for risk of readmission (c) Deep learning classifier model for Tuberculosis images (e) Sleep Apnea prediction. Working on API as service for clinical ML solutions.
- Data Engineering/BigData: A data engineering project which involves multiple ML models for clients, data status monitoring, prediction of delay, prediction of standardisation and processing time, recommendation of resolution. Presently these models are catering to 160+ US clients as these are **agonistic to client data** and **self-learning**. Accuracy of models is **98%** in reduction of false positives. The entire solution is management by using a dashboard designed by our team to monitor performance. Patent is filed for the same.
- Simulations: Developing algorithms for simulating eligible patients for a single registry standard. Developing a Machine learning solution to ensure accurate prediction for qualification of patients (65+ clients).
- Inventory management: (a) Predictive models for delays and reorder status (b) Fraud detection and drug administration models for high valued drugs and a patent for the same is filed in 2021.

Kabbage India Private Ltd, Bangalore

Feb 16, 2018 to Jan 04, 2020

Role: Lead Decision Analyst

Data Science

Responsible for managing **model validation** teams and developing innovative analytical solutions for **credit risk**.

This role involved directly managing a team of 3 data scientists and 2 data analysts. The key projects details are,

- Macro Economics: (a.) Developing automated data extraction engine (b.) Develop Macro indicators crawler for automated data download (c.) Enabling processed macro indicators for risk operations. (d). Working on the development of UI for macro indicators.
- Churn model – Developed a hands-on end to end churn prediction model. This model has a dynamic dependent variable and utilises 4000+ features from 7 different data sources. The model was developed and deployed using pyspark, python and airflow.
- Chatbot – Worked for implementing chatbot for application process. Utilised LDA based clustering and topic modelling techniques.

- Stress Testing – As a lead for a stress testing project was responsible for quantitative and qualitative validation of the LGD model based on the Macro Economic Indicator.
- Causality Framework using Bayesian - Developed a process to validate and test credit risk strategies for reliability using Bayesian Network. Framework was validated on credit scoring.
- Model Monitoring and Validation – Guided the team for the new model monitoring and validation process. Implemented distributed data scraping resulting in higher efficiency and lower turnaround time.
- Response model: Developed hands-on response model for prospective customers to increase originations. A ML model was developed with 72% recall for a class imbalance of 8:10000.
- LSTM deep learning model - developed a transaction classification model using LSTM. The credit transactions are based on embedding learned using Golve-100D.

Behead Pvt Ltd, Bangalore India

Mar 2017 to Jan 2018

Role: Data Scientist

Data Science and Analytics

Data scientist worked with a team of 5 developers and 3 BI solution designers. Some key projects include,

- Artificial Intelligence machine - First the AI engine, dynamic web data is collected using web-scraping; the data collected is image, pdf, and text. Second NLP is performed for topic detection and parts of speech extraction; NLP is done using multilevel parameter estimation using exponential growth. Third classification is performed using deep learning model.
- Client based data science models developed for following use-cases, (a) Gap analysis, (b) Emerging market analysis, and (c) Monthly/Quarterly sales prediction model.

Indian Institute of Science, Bangalore

July 2013 to Feb 2017

Research scholar of Doctoral programme for Engineering

Analytics and Data Science

Thesis weblink - <https://etd.iisc.ac.in/handle/2005/4096>

Explores better understanding of Retail online customer decision making, in low involvement product categories (groceries), with special focus on identification of consumer cognition factors which play a significant role in online purchase intention. Specifically, we look at, (a) patterns of online purchase in grocery purchase orders; (b) differences in purchase and non-purchase behaviour; (c) online purchase behaviour of each grocery category; and (d) effects of cognitive attributes on online grocery purchase behaviour.

- Data is collected through a formal collaboration with a leading online grocery store ([BigBasket](#)) in India. Multi-method research design is used, and raw weblog files (messy data) is cleaned/prepared before analysis. A new data cleaning algorithm was developed for messy data.
- Data science: CART decision trees, Neural network, Association rules, Bayesian Network, Regression, and significance tests were utilized.

Dayananda Sagar Business Academy, Bangalore India

Jun 2012 to July 2013

Lecturer for MBA

Mathematics and Statistics

Hewlett-Packard Global Soft Pvt Ltd, Bangalore India

Nov 2008 to Sep 2010

Technical Support SAP Servers

Incident Co-ordinator

TECHNICAL SKILLS

Cloud	Programming	Database	Data Science
GCP, AZURE, DATABRICKS, AGILE, SLDC, LLD approaches	○ R- programming ○ Python ○ Basics PySpark,	○ Kafka ○ SQL MYSQL ○ Visualisation using Power BI and Tableau	○ ML and Deep Learning ○ Time series ○ GPT models, LLM ○ NLP text based models ○ Image processing by GENAI

PATENTS

1. "Intelligent System for Automatically Testing and Selecting from Multiple Data Models for Accurate Diversion Prediction". date: Oct 9, 2021 ; country patent filed: USA ; Patent Number: US20230077660A1
weblink- <https://patents.justia.com/patent/20230077660>
2. "Active management of files being processed in enterprise data warehouses utilizing time series active management of files being processed in enterprise data warehouses utilizing time series". date: Jan 30, 2023 ; country patent filed: USA; Patent Number: US12111848B2
weblink: <https://patents.justia.com/patent/12111848>
3. "Outreach communication controls using machine learning"; File.no: 65/583,157 ; date: Sep 15, 2023 ; country patent filed: USA
4. "Obstructive sleep apnea prediction and analytical reasoning using hyper parameters for accurate modelling of risk" ; File.no: 63/624734; date: Jan 24, 2024 ; country patent filed: USA

QUALIFICATION

- Doctorate (PhD) in Engineering (Consumer Analytics) from [Indian Institute of Science](#), Bangalore in 2018.
- Master's in Business Administration (MBA) from [Central University of Karnataka](#), Gulbarga in June 2012
- Bachelor's in computer science engineering (BE/BTECH), [Visveshwaraya Technological University](#), Belgaum in 2008.

PERSONAL DETAILS

Date of Birth: 01 May 1987 | Marital Status: Married | Nationality: Indian

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